BTS-VII-11.18-1871 Reg. No. [ ]

C

B.Tech. Degree VII Semester Examination November 2018

ME 15-1703 MACHINE DESIGN II

(2015 Scheme)

(Use of Design Data Handbook is permitted. Data not given may be suitably assumed. All such assumptions must be clearly stated.)

Time: 3 Hours Maximum Marks: 60

PART A

(Answer *ALL* questions)

(10 × 2 = 20)

I. (a) Differentiate between single plate and multi plate clutches with application.

(b) Explain the effect of centrifugal tension in belt on power transmission.

(c) Explain the working of band brakes.

(d) What are the points to be considered while designing a sliding mesh type of multi-speed gear box?

(e) Define "formative number of teeth" as applied to helical gears and explain its importance in the design of helical gears.

(f) Explain the theory of hydrodynamic lubrication with neat sketch.

(g) What is rating life of a rolling contact bearing? Explain it with an example.

(h) Explain the significance of the bearing characteristic number in the design of sliding contact bearing.

(i) What are the main design considerations for the parts produced in milling machines?

(j) Describe about tolerance and surface finish.

PART B

(4 × 10 = 40)

II. Design a cone clutch to transmit 40 kW at 750 rpm. Also determine the axial force required to transmit the torque and axial force required to engage the clutch. Assume $f = 0.4$ and $p = 0.2$ N/mm^2 for friction material ($\alpha = 12.5°$). Take $D_m/b = 6$, where D_m is the mean diameter of friction lining and b is the face width. (10)

OR

III. Select a V-belt drive to connect a 15 kW, 2880 rpm motor to a centrifugal pump, running at approximately 2400 rpm, for a service of 18 hours per day. The centre distance should be approximately 400 mm. Assume the pitch diameter of driving pulley as 125 mm. (10)

IV. A 12 kW motor running at 1170 rpm drives a fan through a pair of spur gears (forged steel SAE 1030 pinion and CI gear) with a reduction ratio of 3:9:1. Design the gear and check for dynamic and wear loads. (10)

OR

V. The vertical spindle of a drilling machine is to be driven by a pair of straight bevel gears with 20° involute teeth. The speed reduction is 4:1. The drill requires a power of 50 kW at 720 rpm. A service factor of 1.35 may be taken, and choose the steel (C30) as material for both gear and pinion. Design the gear pair. (10)

(P.T.O.)

VI. It is required to design a main bearing of a 4-stroke oil engine to sustain a load of 6 kN over a shaft of diameter 50 mm. The operating speed of the shaft is 1000 rpm and the operating temperature is $50°$ C. Determine: (10)

(i) Dimensions of the bearing
(ii) The viscosity of the oil to be used for the bearing and hence suggest appropriate oil
(iii) Coefficient of friction
(iv) Heat generated
(v) Heat dissipated
(vi) Heat to be removed by artificial cooling if necessary
(vii) Sommerfeld number

OR

VII. A ball bearing supporting a bevel gear shaft is subjected to a radial load of 4000 N and a thrust load of 5000 N. Select a proper bearing for a speed of 1600 rpm. Life expectancy is 5 years at 10 hours per day. (10)

VIII. What are the general design recommendations for (i) rolled sections (ii) welded parts? Explain with sketches. (10)

OR

IX. (a) Explain Indian Standard System of limits and fits. (5)

(b) What are the general design recommendations for castings? (5)

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